STAT 5701 **Midterm Exam** **Name: Solutions**

Show your work for all parts!

Fall 2025 **Uni:**

1. Let X and Y be two continuous random variables with a joint pdf given by

$$f(x,y) = cx^2y(1+y), \qquad 0 \le x \le 3, \qquad 0 \le y \le 3$$

a) (6 p) Find the value of c that makes this a valid density function.

$$\int_0^3 \int_0^3 cx^2y(1+y)dxdy = c\int_0^3 y(1+y)\frac{x^3}{3}\Big|_0^3 dy = 9c\int_0^3 (y+y^2)dy$$

$$= 9c\left(\frac{y^2}{2}+\frac{y^3}{3}\right)\Big|_0^3 = 9c\left(\frac{9}{2}+9\right) = 1 \Leftrightarrow \frac{9c(27)}{2} = 1$$

$$\Rightarrow c = \frac{2}{243}$$

b) (6 p) Find the marginal pdf $f_X(x)$.

$$f_X(x) = \frac{2}{243}x^2\int_0^3 y(1+y)dy = \frac{2x^2}{243}\left(\frac{y^2}{2}+\frac{y^3}{3}\right)\Big|_0^3$$

$$= \frac{2x^2}{243}\left(\frac{9}{2}+9\right) = \frac{2x^2}{243}\left(\frac{27}{2}\right)$$

$$= \frac{x^2}{9}, x \in [0,3]$$

2. Let the probability density function of a continuous r.v. X be given by

$$f(x) = \frac{c}{x^3}, \qquad x \geq 1$$

a. (6 p) Find c. (If you can't find c, assume $c = 1$ and move on.)

$$\int_1^\infty \frac{c}{x^3} dx = 1 \iff -\frac{c}{2} x^{-2} \Big|_1^\infty = 1$$
$$\iff -\frac{c}{2}(0 - 1) = 1 \iff \frac{c}{2} = 1 \iff c = 2$$

b. (6 p) Find the cumulative distribution function $F(x)$.

$$F(x) = \int_1^x \frac{2}{t^3} dt = -t^{-2}|_1^x = 1 - \frac{1}{x^2}, x \geq 1$$

c. (5 p) Find P(2 < X < 3).

$$\mathrm{P}(2 < X < 3) = \mathrm{F}(3) - \mathrm{F}(2) = \left(1 - \frac{1}{3^2}\right) - \left(1 - \frac{1}{2^2}\right) = \frac{8}{9} - \frac{3}{4} = \frac{5}{36}$$

d. (+5 p) **Bonus**: Does the moment generating function of X exist? Explain why or why not.

Does not exist because not all moments exist.

3. A fair coin is tossed *four* times, and the resulting sequence of heads and tails is recorded. Define the following events:

A = {exactly two heads appear}
B = {heads and tails alternate}
C = {the last toss is tails}

Assume tosses are independent and each toss is equally likely to be tails or heads. Find the following probabilities:

a. (5 p) $P(B \cap \bar{C})$.

The sample space has $2^4 = 16$ outcomes.
A = {HHTT, HTHT, HTTH, THHT, TTHH, THTH}
B = {HTHT, THTH} and C = {HHHT, HHTT, HTHT, THHT, TTHT, THTT, HTTT, TTTT}.
$\bar{C}$ = {HHHH, HHTH, HTHH, THHH, TTHH, THTH, HTTH, TTTH}

$$\Rightarrow B \cap \bar{C} = \{THTH\}$$
$$\Rightarrow P(B \cap \bar{C}) = \frac{1}{16}$$

b. (5 p) P($A \mid C$).

$A \cap C$ = {HTHT, HHTT, THHT}

$$\Rightarrow P(A \cap C) = \frac{3}{16}$$

$$\Rightarrow P(A|C) = \frac{P(A \cap C)}{P(C)} = \frac{\frac{3}{16}}{\frac{8}{16}} = \frac{3}{8}$$

c. (5 p) $P(\bar{B} \cup \bar{C})$.

By De Morgan's law, $\bar{B} \cup \bar{C} = \overline{B \cap C}$

$$\Rightarrow P(\bar{B} \cup \bar{C}) = 1 - P(B \cap C) = 1 - \frac{1}{16} = \frac{15}{16}$$

4. Let X and Y be two discrete random variables whose joint and marginal distributions are partially given in the following table:

$X \setminus Y$	1	2	3	$p(x)$
1	1/6	0	1/6	1/3
2	0	1/4	1/12	1/3
3	0	1/12	1/4	1/3
$p(y)$	1/6	1/3	1/2	

a) (6 p) Complete the table.

b) (5 p) Find P($X = Y$)

$$\mathrm{P}(X = Y) = \frac{1}{6} + \frac{1}{4} + \frac{1}{4} = \frac{2}{3}$$

c) (5 p) Find E(Y).

$$E(Y) = 1 \times \frac{1}{6} + 2 \times \frac{1}{3} + 3 \times \frac{1}{2} = \frac{7}{3}$$

d) (5 p) Determine the conditional pmf $p(x \mid y = 2)$.

x	1	2	3
$p(x \mid y = 2)$	0	3/4	1/4

5. A *discrete* random variable X has the following probability distribution

x	1	2	3	4
$p(x)$	0.5	0.2	0.1	0.2

a. (6 p) Find the cumulative distribution function $F(x)$.

$$F(x) = \begin{cases} 0, & x < 1 \\ 0.5, & x \in [1,2) \\ 0.7, & x \in [2,3) \\ 0.8, & x \in [3,4) \\ 1, & x \geq 4 \end{cases}$$

b. (6 p) Find Var(X).

$$E(X) = 1(0.5) + 2(0.2) + 3(0.1) + 4(0.2) = 2$$

$$E(X^2) = 1^2(0.5) + 2^2(0.2) + 3^2(0.1) + 4^2(0.2) = 5.4$$

$$Var(X) = 5.4 - 2^2 = 1.4$$

c. (5 p) Find the mgf $m_X(t)$.

$$m_X(t) = 0.5e^t + 0.2e^{2t} + 0.1e^{3t} + 0.2e^{4t}$$

6. A message is transmitted as a binary code of 0s and 1s. Experience with the equipment being used suggests that if a 0 is sent, it will be received as 0 94% of the time. If a 1 is sent, it will be received as a 1 95% of the time (and as a 0 5% of the time). The text being sent is known to be 7% 1s and 93% 0s.

   a. (6 p) What is the probability that the received message is 1?

Let R_i be the event that a message is received as i, where i =0 or 1
Let S_i be the event that a message is sent as i, where i =0 or 1
Then

$$P(R_1) = P(S_0)P(R_1|S_0) + P(S_1)P(R_1|S_1)$$
$$= 0.93(1 - 0.94) + 0.07(0.95) = 0.1223$$

   b. (6 p) Suppose the next signal sent is received as a 1. What is the probability it was sent as a 0?

$$P(S_0|R_1) = \frac{P(S_0)P(R_1|S_0)}{P(R_1)}$$
$$= \frac{0.93(1 - 0.94)}{0.1223} = 0.456$$

   c. (6 p) If the *same* message is sent twice, and assuming the received messages are conditionally independent, what is the probability that it is received as two 1's?

$$P(R_1 \cap R_1) = P(S_1)P(R_1 \cap R_1|S_1) + P(S_0)P(R_1 \cap R_1|S_0)$$
$$= 0.07(0.95)^2 + 0.93(0.06)^2 = 0.066523$$